

Statistics — MCQ Worksheet

Mathematics • Chapter 14 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. The mean of n observations $x_1, x_2, \dots, x_n$ is:

- (A) $(x_1 + x_2 + \dots + x_n)/n$
(C) Largest value

- (B) $x_1 \times x_2 \times \dots \times x_n$
(D) $\text{Sum} \times n$

Q2. The mean of grouped data using direct method is:

- (A) $\Sigma(f \cdot x)/\Sigma f$
(C) Σf

- (B) $\Sigma x/n$
(D) Mode

Q3. In assumed mean method, the mean is:

- (A) $a + (\Sigma f \cdot d/\Sigma f)$
(C) $a - \Sigma f \cdot d$

- (B) $a \times \Sigma f \cdot d$
(D) $\Sigma d/n$

Q4. In step-deviation method d becomes:

- (A) $x - a$
(C) $(x - a)/h$

- (B) $x + a$
(D) $(x + a)/h$

Q5. Median is the value that lies in the middle when data is arranged in:

- (A) Random order
(C) No order

- (B) Ascending or descending order
(D) Reverse order

Q6. Mode is the value with the:

- (A) Highest frequency
(C) Middle frequency

- (B) Lowest frequency
(D) Random frequency

Q7. Empirical relation between mean, median and mode is:

- (A) $\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$
(C) $\text{Mean} = \text{Mode} + \text{Median}$

- (B) $\text{Mode} = 2 \text{ Mean} - \text{Median}$
(D) $\text{Mode} = \text{Mean} - \text{Median}$

Q8. For grouped data, $\text{median} = l + ((n/2 - cf)/f) \times h$ where l is:

- (A) Lower limit of median class
(C) Mode

- (B) Mean
(D) Upper limit

Q9. For grouped data, $\text{mode} = l + ((f_1 - f_0)/(2f_1 - f_0 - f_2)) \times h$ where f_1 is:

- (A) Frequency of preceding class
(C) Frequency of succeeding class

- (B) Frequency of modal class
(D) Cumulative frequency

Q10. Mean of first 10 natural numbers is:

- (A) 5
(C) 10

- (B) 5.5
(D) 11

Q11. Mean of 5, 10, 15, 20, 25 is:

- (A) 10
(C) 20

- (B) 15
(D) 25

Q12. Median of 11, 12, 14, 13, 15 (arranged) is:

- (A) 11
(C) 13

- (B) 12
(D) 15

Q13. For an even number of observations, median is the:

- (A) Largest
(C) Average of two middle values

- (B) Smallest
(D) Mode

Q14. Mode of 2, 3, 5, 5, 7, 8 is:

- (A) 2
(C) 5

- (B) 3
(D) 7

Q15. In a cumulative frequency curve (ogive), the median can be found by drawing a line from cumulative frequency value of:

- (A) n
- (B) $n/2$
- (C) $n/4$
- (D) $2n$

Q16. Less-than ogive plots:

- (A) Upper limits vs cumulative frequency
- (B) Lower limits vs cumulative frequency
- (C) Midpoints vs frequency
- (D) Frequency vs class

Q17. More-than ogive plots:

- (A) Upper limits vs cumulative frequency
- (B) Lower limits vs cumulative frequency
- (C) Midpoints vs frequency
- (D) Frequency vs class

Q18. The middlemost value of a data is its:

- (A) Mean
- (B) Median
- (C) Mode
- (D) Range

Q19. Class mark of class 10–20 is:

- (A) 10
- (B) 20
- (C) 15
- (D) 30

Q20. If the class size is h and lower class boundary l , the class mark equals:

- (A) $l + h/2$
- (B) $l - h$
- (C) $l \times h$
- (D) $2l + h$

Q21. Mean is most affected by:

- (A) Mode
- (B) Extreme values (outliers)
- (C) Median
- (D) Range

Q22. In a moderately skewed distribution, $\text{mean} - \text{mode} \approx$

- (A) $3(\text{mean} - \text{median})$
- (B) $2(\text{mean} - \text{median})$
- (C) $\text{mean} - \text{median}$
- (D) 0

Q23. Σf in grouped data represents:

- (A) Total frequency
- (B) Sum of x 's
- (C) Mean
- (D) Mode

Q24. If the mean of 5, 7, 9, x is 8, then $x =$

- (A) 10
- (B) 11
- (C) 12
- (D) 9

Q25. For a symmetric distribution $\text{mean} = \text{median} = \text{mode}$. This is true:

- (A) Always
- (B) Only sometimes
- (C) Never
- (D) Only for skewed